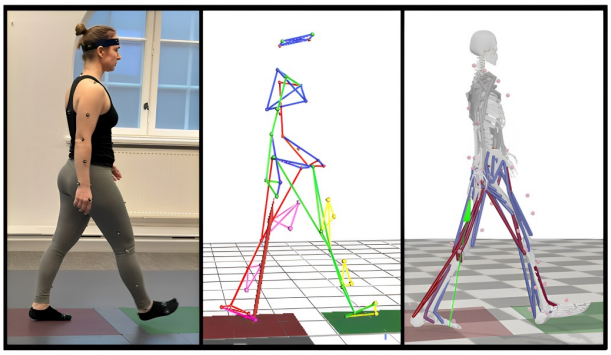
SM2501 /FSG3084

Biomechanics of Human Movement

Lecture 1

- Introduction
- Terminology
- Skeletal system and bone tissue



1

Instructor:

Lanie Gutierrez-Farewik
Professor of Biomechanics KTH (affil. KI Women's & Children's Health)
Director KTH MoveAbility, Dept. Engineering Mechanics, Osquars Backe 18

BSc Mechanical Engineering (Cornell Univ)
MSc Biomedical Engineering (Univ of Michigan)
PhD Orthopedics (Karolinska Inst)
Clinical gait analysis engineer, Karolinska Univ Hospital (consultant)

Assistant teacher:
Ruoli Wang, Associate Professor, KTH MoveAbility

Course assistant/TA:
Jennifer Maier, PhD, Post doc, KTH MoveAbility

Guest lecturers:
Stefan Gantelius MD PhD, Head of Dept. of Orthopedics, Danderyd Hospital
Toni Arndt, Professor, GIH

2

This week

Anatomical terminology

Bones – skeletal system and bone tissue

Soft tissues

Muscle functional anatomy and tissue

Data collection at **Promobilia MoveAbility Lab**

Osquars Backe 18, 4th floor

Entrance just across from KTH Library

3

Schedule

Course PM

Participants

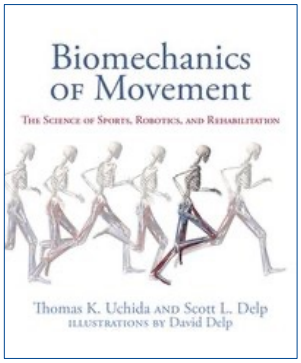
4

Course PM, book, software

Biomechanics of Movement

T Uchida and S Delp, MIT Press 2021

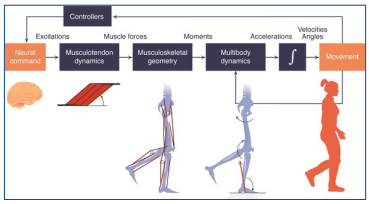
Available online at KTH Library



OpenSIM

Version 4.5

https://simtk.org/projects/opensim



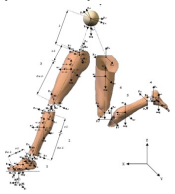
Series from author S Delp:

https://www.youtube.com/channel/UCDNGy0KKNLQ-ztcL5h2Z6zA

5

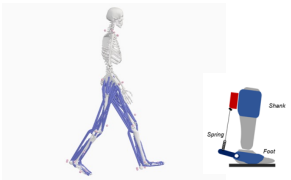
Group project assignments (max 3 students/group)

- Proj 1 (Matlab)



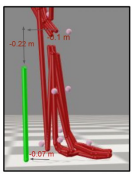
- Computation and interpretation of body movement, joint moments and powers from collected data

- Proj 2-3 (OpenSim)



- Simulation and interpretation of movement, kinetics, muscle activation and forces
- Simulation-based design of ankle exoskeleton

- Final project (OpenSim)



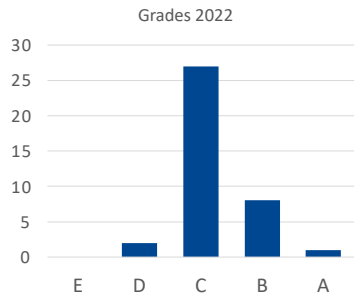
- Simulation of altered muscle strength, interpretation of compensation
- Simulation of ankle exoskeleton as compensation

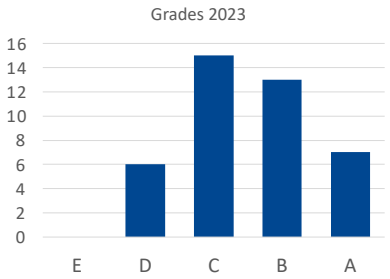
All: Discussion and critical evaluation of methods and results

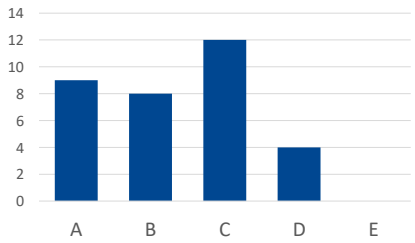
6

Results and evaluation 2022-4

(uploaded to Course web)







7

Note about Thursday's lab

Compulsory!

Three sessions. I will arrange a sign-up on Canvas tomorrow.

9.15 – 10.00

10.15 – 11.00

11.15 – 12.00

We need 1 volunteer for the entire time (8.45 – 12.05)

8

What is biomechanics?

DeLuca Foundation

https://www.youtube.com/watch?v=OhN1d0xxUjE

9

Biomechanics jobs

Nanosciences

Automobile/crash safety

Orthopedic product development

Robotics

Veterinary sciences

Academia

Biofluids research and product design

Clinical motion analysis

Sports research

Product/software support

Tissue engineering

Animation

Clinical research

Ergonomics

Evolutionary biomechanics

Plant biomechanics

Sports product development

Rehabilitation/assistive devices

Accident reconstruction / insurance

10

Case

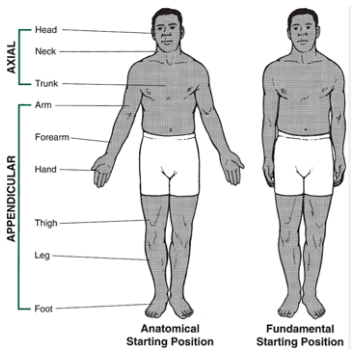


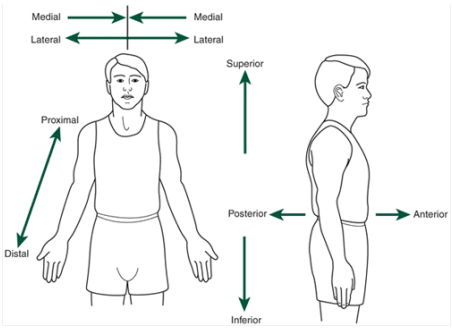
Describe his walking

Why does he walk like this?
'Cause' and 'effect'?
What should one treat? How?
What should we expect?
Is it better? Good enough?

11

Terminology

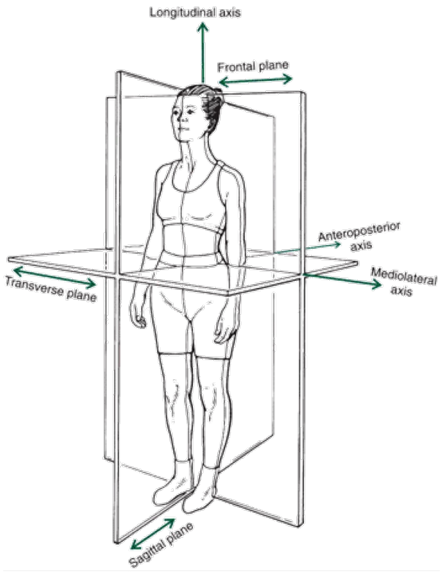




Hamill & Knutzen 2009

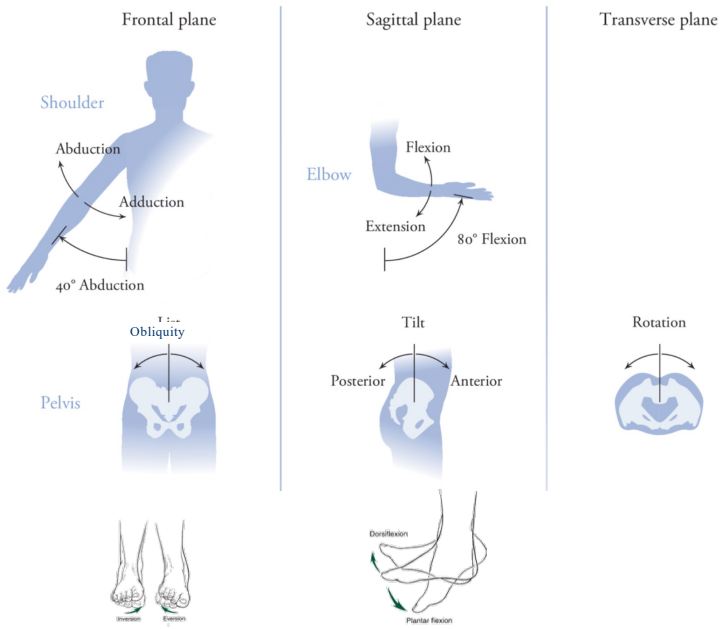
12

Anatomical planes



Hamill & Knutzen 2009

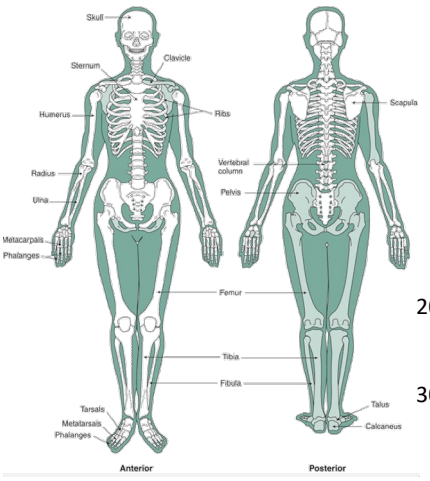
13



Uchida & Delp 2021, Hamill & Knutzen 2009

14

Skeletal system



206 adults
52 feet
54 hands
300-350 infants

Functions

- Leverage*
- Support*
- Protection
- Storage
- Blood cell formation

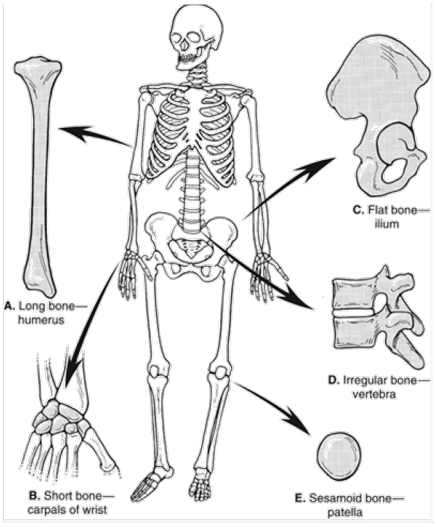
*Levers and support

- Simple machine
- Magnifies force and/or speed of movement
- Morphology: Shape and structural arrangement of bones and characteristics of the articulations connecting the bones

Hamill & Knutzen 2009

15

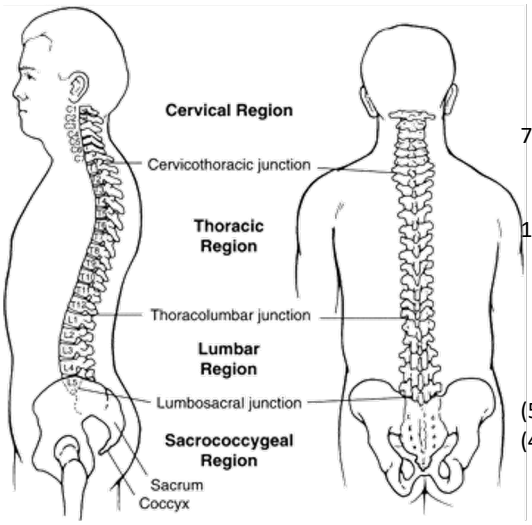
Types of bones



Hamill & Knutzen 2009

16

Spine and vertebra



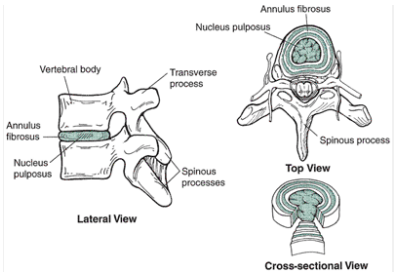
7 Cervical

12 Thoracic

5 Lumbar

(5) Sacral

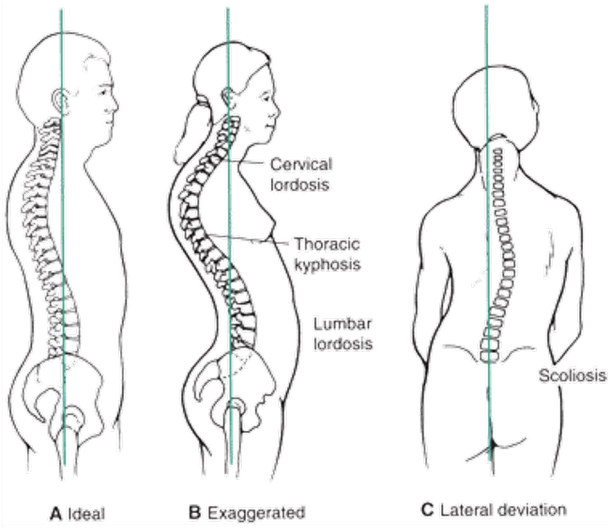
(4-5) Coccygeal



Hamill & Knutzen 2009

17

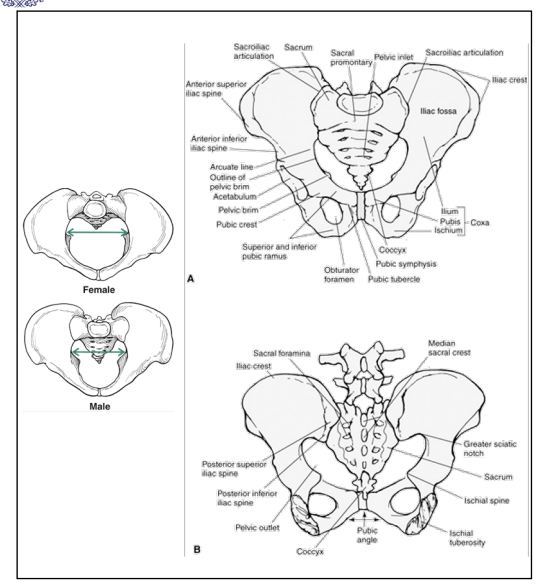
Pathological back postures



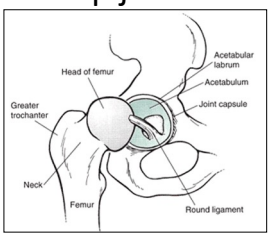
Hamill & Knutzen 2009

18

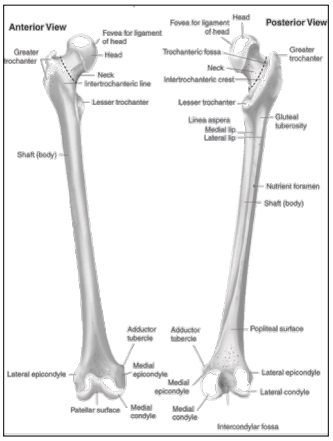
Pelvis



Hip joint



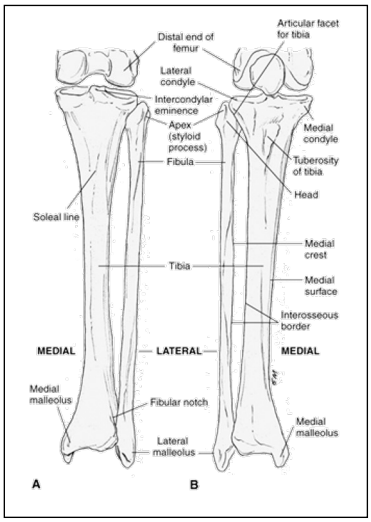
Femur



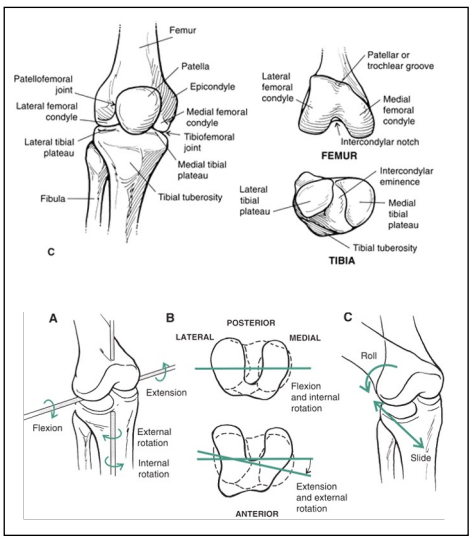
Hamill & Knutzen 2009

19

Tibia + Fibula



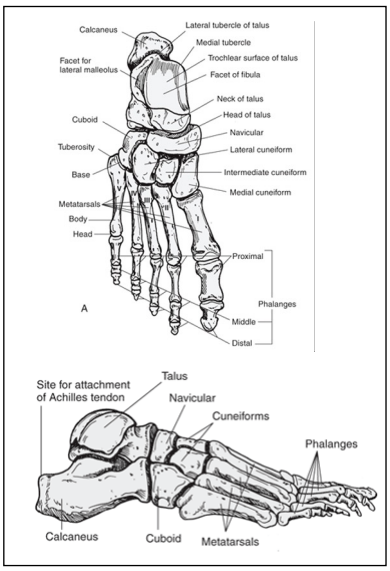
Knee Joint



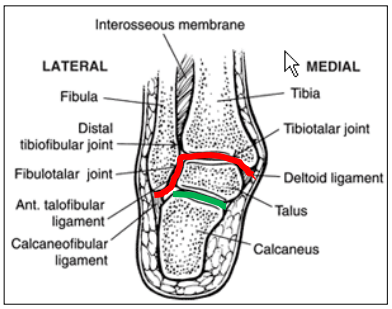
Hamill & Knutzen 2009

20

Foot



Ankle joint(s)



Talocrural joint
Subtalar joint

Hamill & Knutzen 2009

21

Bone (osseus) tissue, characteristics and content

- Light
- High tensile and compressive strength
- Relatively high elasticity
- Dynamic
- 5-7% recycled every week
- Surrounded by periosteum: dense fibrous membrane, permeated by blood vessels
- Approx. 20% of total body weight
- Influenced by nutrition, physical activity, and postural habits

Matrix of

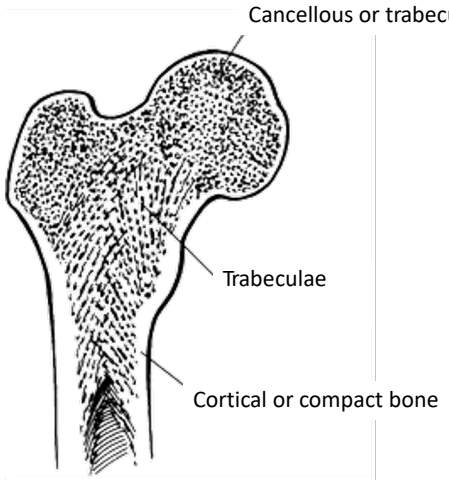
- Inorganic minerals (60-70% of dry weight)
  - Calcium and phosphate
  - Compression strength and rigidity
- Collagen (25-30% dry weight)
  - Found in all connective tissue
  - Flexibility and resiliance
- Water up to 25% total weight (85% in organic matrix, rest in canals and cavities)

Osteocytes: bone cells

- Osteoblasts: create bone. Produce organic fibers on which calcium is deposited
- Osteoclasts: resorb bone: Break down bone, make calcium soluble

22

Bone tissue type

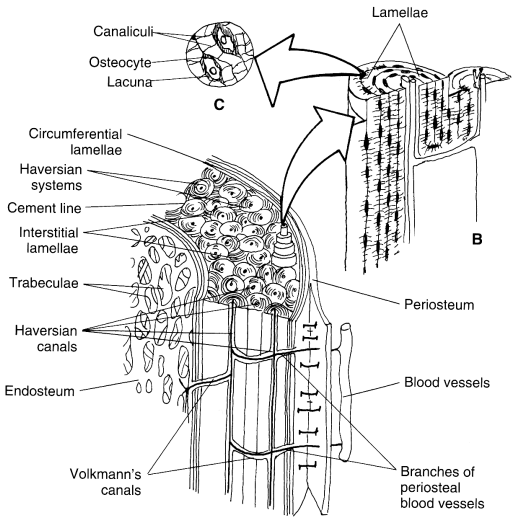


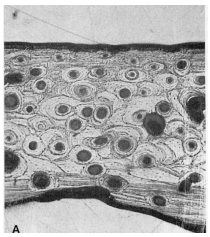
Nordin & Frankel 2001

23

Cortical bone

Osteons, Haversian systems





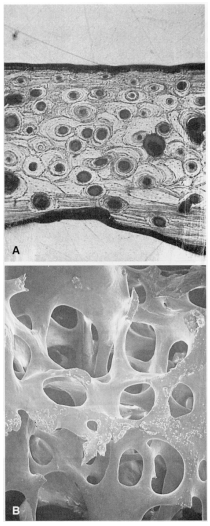
Cross section

Nordin & Frankel 2001

24

Cancellous/Spongy/Trabecular bone

- Thin trabeculae in loose mesh
  - Marrow in spaces between
- Always surrounded by cortical bone



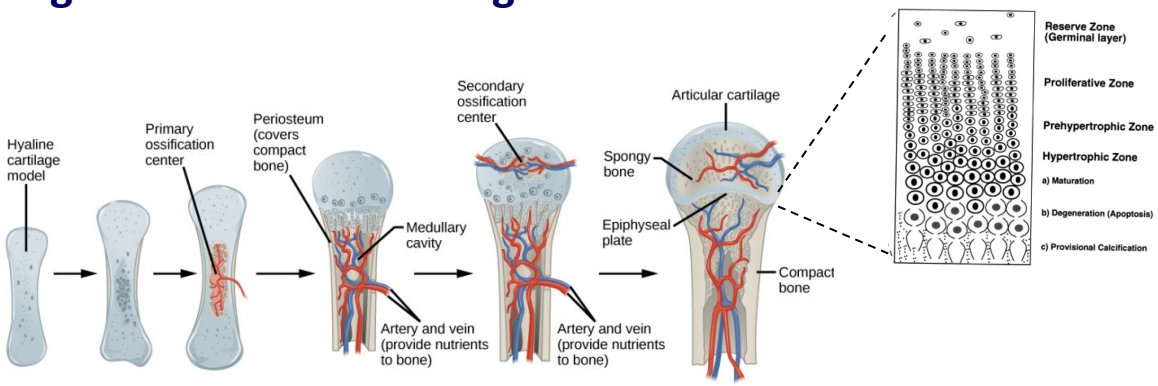
Cortical

Trabecular

Nordin & Frankel 2001

25

Long bone formation and growth



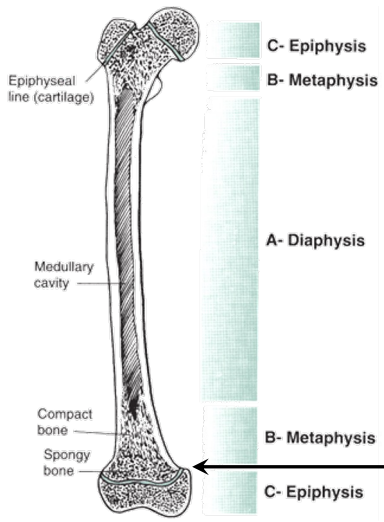
Endochondral ossification

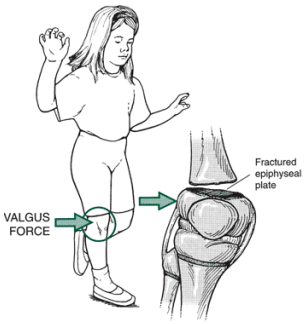
Molnar & Gair, Concepts of Biology, 1st Canadian Ed.

Stevens and Williams, Mollec. Cell Endocrin, 1999

26

Long bones, ex. femur

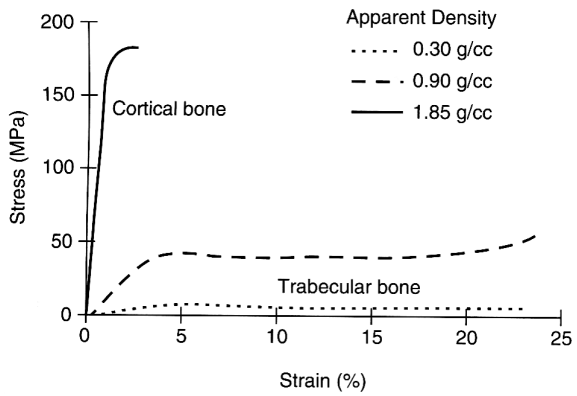




Hamill & Knutzen 2009

27

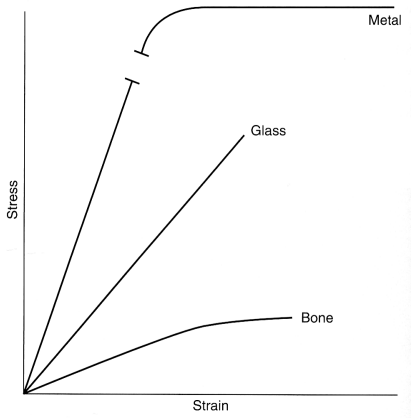
Bone material properties: Stiffness in compression

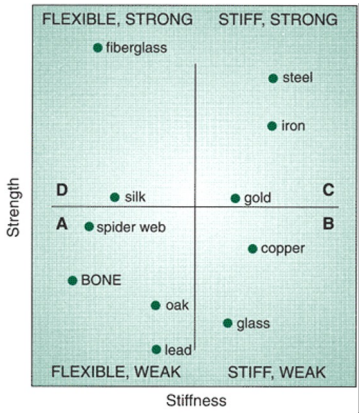


Nordin & Frankel 2001

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Bone material properties: Relative to other materials



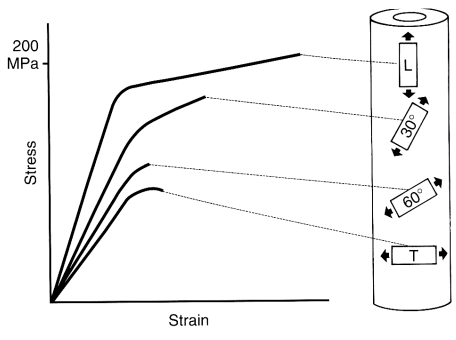


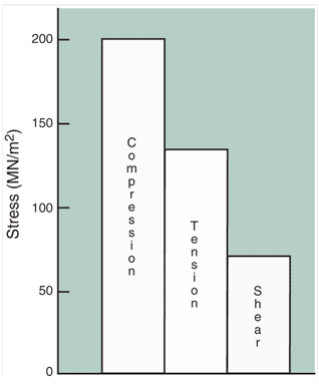
Nordin & Frankel 2001

Hamill & Knutzen 2009

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Bone material properties: Anisotropy



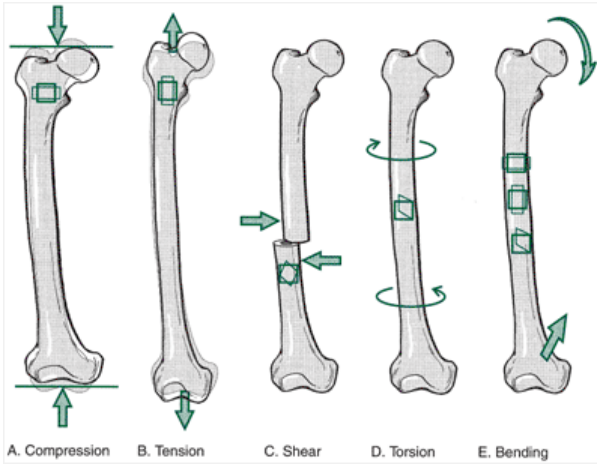


Nordin & Frankel 2001

Hamill & Knutzen 2009

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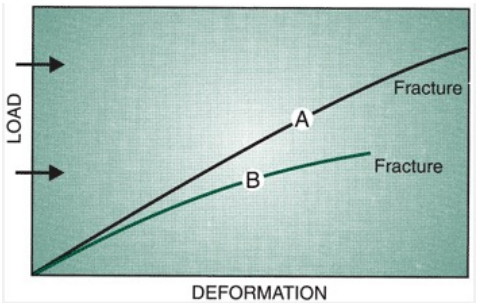
Modes of loading and tissue stress



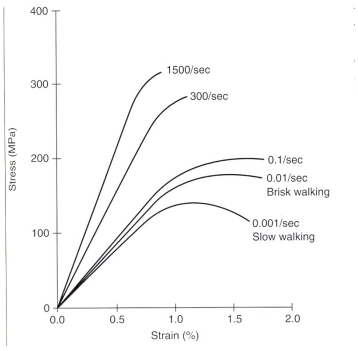
Hamill & Knutzen 2009

31

Bone material properties: Viscoelasticity



A: fast loading
B: slow loading



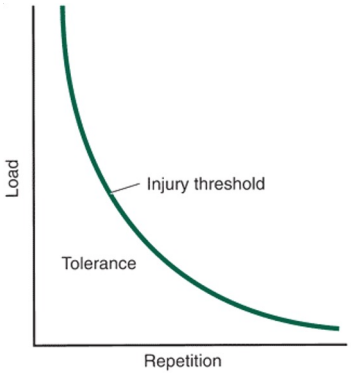
Hamill & Knutzen 2009

Nordin & Frankel 2001

32

Strength and failure (General)

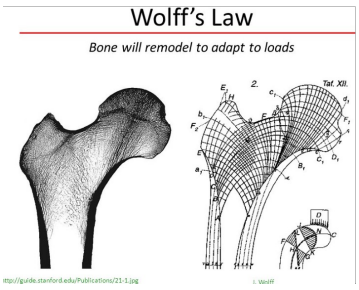
- Failure point or load sustained before failure
- Failure is caused by:
  - Single traumatic event
  - Accumulation of microfractures
  - 10% of injuries to athletes are stress fractures
- Assessed by:
  - Energy storage
  - Area under stress-strain curve

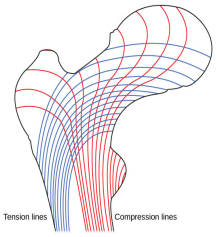


33

Wolff's Law

- Julius Wolff, German anatomist
- Resorption
  - Response to decreased stress
  - Osteoclasts dominate
    - Disuse, immobilization, microgravity
- Deposition
  - Response to increased stress
  - Osteoblasts dominate
    - Weight-bearing exercise





Molnar & Gair, Concepts of Biology, 1st Canadian Ed.

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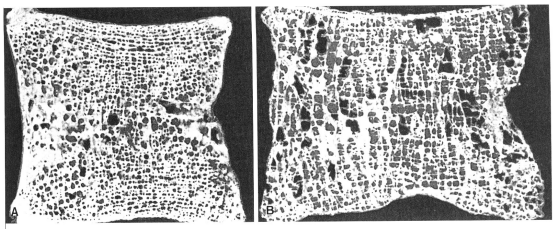
Bones & Physical Activity

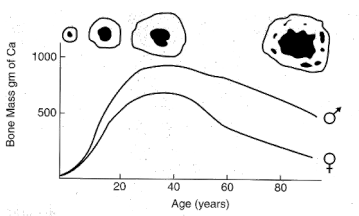
- Bones require mechanical stress to grow and strengthen
  - Dynamic loading better than static
  - Higher frequencies more effective
  - Shorter, repeated loading periods better than fewer longer periods
- Loading → Deposition → Increased Density

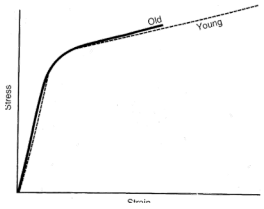
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Aging, gender

Young bone Old bone





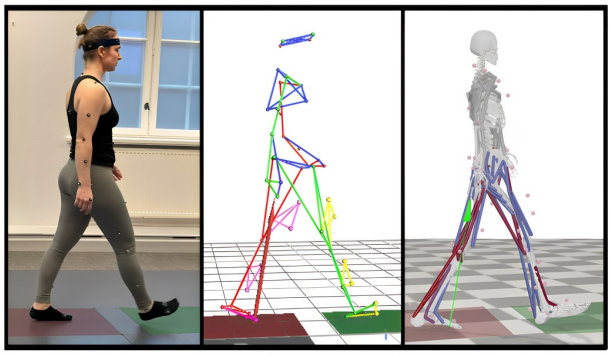


Nordin & Frankel 2001

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For next time:

- OpenSIM – own PCs/Macs?
- Read!
- Volunteer for Thursday?



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